

The Trajectory Before the Statistic

Finite Symbolic Trajectories, Entropy, and the Poincaré
Reconstruction of Measured Results

A Monograph in Finite Symbolic Mechanics

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September 2026

Before a result became statistical, it was a trajectory. Before a mathematical method became a noun, it too was a trajectory. This monograph begins where both trajectories disappear.

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Preface: Before the Numbers Became Statistics

This inquiry began with two small words: *probit* and *logit*. Both now appear as ordinary objects within statistics. They can be placed in a sentence, selected from a software menu, taught in a course, and applied to a set of registered values without requiring the user to replay the historical operations from which either word arose. This is useful. No language could function if every utterance had to carry the whole trajectory of its formation. Yet the efficiency of the noun can also conceal the fact that there was first a problem, an inherited language, a series of measurements, a computational constraint, a competition among procedures, a selection, and only later a name. What looks stationary at the end was moving throughout its formation.

The same compression occurs on the other side of the statistical operation. A measured process produces an ordered sequence of registrations. Each registration follows another; each has an interval, an instrumental condition, a threshold, a resolution, and a position within a history. The sequence is not initially a set of interchangeable numbers. It becomes such a set only when a handler is licensed to disregard some of the relations that made it a trajectory. A mean retains one relation, a histogram another, a regression another, and an autoregressive model yet another. Each statistical construction preserves something, but each also determines which differences between trajectories will no longer count within the result.

The argument of this monograph is therefore not that statistics is false, nor that compression should be abandoned. A finite symbolic practice cannot retain every possible relation in every act of communication. The argument is that the compression should remain visible as an operation. When it disappears, the statistical result can be mistaken for the measured process, just as the mathematical noun can be mistaken for the historical operation from which it descended. Finite Symbolic Mechanics, abbreviated throughout as FSM, places the operation back into the account. It asks what was registered, what order was retained, which handler acted upon the registrations, which distinctions survived, and which formerly distinguishable trajectories were placed beneath the same symbol.

This route gives entropy a particular position. Entropy will not be introduced as a substance residing inside the numbers or as a synonym for disorder. It will be approached as unresolved multiplicity relative to a finite symbolic registration. Once several distinguishable trajectories have been assigned the same surviving result, that result cannot by itself determine which trajectory occurred. The resulting uncertainty is not necessarily uncertainty in the original act of measurement. It may have been generated later by the symbolic handling of registrations that were individually precise. Measurement uncertainty and trajectory uncertainty must therefore be separated.

The monograph then turns from compression to reconstruction. Where the ordered registrations have been preserved, a Functional Symbolic Trajectory can be embedded using delayed relations. A Poincaré section can then be constructed through the resulting geometry and a return map formed from successive crossings. In FSM this is not a metaphor

for sampling. It is an operational procedure performed upon a finite symbolic trajectory. The procedure cannot restore relations already destroyed, and it cannot recover an inaccessible exact past. It can, however, expose dynamical relations that remain carried by the ordered registrations and which a distribution or aggregate may not reveal.

The route through the monograph is consequently circular in the productive sense. It begins with a named statistical object, follows the lost historical trajectory beneath the name, moves to the measured trajectory beneath the result, identifies the handler that contracts many trajectories into one statistical registration, and then returns to the retained sequence in order to reconstruct its relational geometry. The statistic is not removed. It is placed downstream of the trajectory from which it was made.

A Note on Formal Symbols and the Equals Sign

FSM cannot examine symbolic authority while quietly granting its own notation an unlimited authority. The formal expressions used here are therefore local declarations within stated operations. When an expression such as

$$\mathcal{C}(\mathcal{T}_A) = \mathcal{C}(\mathcal{T}_B) \tag{1}$$

is written, the equals sign does not declare that the two physical histories are identical, nor that the trajectories are interchangeable under every possible investigation. It says that the specified handler $\mathcal{C}$ has returned the same registered output for both trajectories within the declared symbolic procedure. Equation (1) is an operational output identity, not an ontological identity.

Definitions will often use the symbol $:=$. This marks a declared construction rather than the discovery of a timeless identity. The notation

$$\mathcal{T} := (r_1, r_2, \dots, r_N) \tag{2}$$

states that, for the present work, the symbol $\mathcal{T}$ will carry the ordered finite sequence of registrations r_1 through r_N . It does not remove the physical production of those registrations, their uncertainty, or their provenance. The notation is a handle placed upon a trajectory so that further operations can be described without repeatedly reproducing the entire account.

This local discipline applies equally to probability, entropy, geometry, and reconstruction. Every variable will be tied to a finite symbolic role. A probability weighting will require an admissible family and a rule by which its weights were produced. An entropy will require a surviving registration and the unresolved alternatives compatible with it. A reconstructed state will require an ordered finite sequence and a declared delay operation. A Poincaré return will require a constructed section and registered crossings. The symbols do not arrive before the operations that warrant them.

I The Noun and Its Forgotten Trajectory

Probit, Logit, and the Naming of an Operation

The words *probit* and *logit* appear to a modern reader as names belonging naturally to the landscape of statistics. A student learns that a probit model uses one link and a logit model another. A researcher selects a command, receives coefficients, and reports that the analysis has been performed. The compactness of the words makes the procedure transferable. Yet that compactness is terminal. It presents the end of a trajectory while withholding the route by which the endpoint was reached.

The logistic form entered modern statistics through a history that included population arguments, bounded growth, the practical fitting of curves, the rediscovery of earlier work, and the computational conditions under which one procedure could be preferred to another. The name *logit* did not descend upon an already completed object. It supplied a portable handle for an operation that had become useful within a developing statistical corpus. Once the handle became stable, later users no longer needed to know why that particular operation had survived, which alternatives had competed with it, or which burdens of hand calculation had influenced its adoption.

This is not a defect peculiar to statistics. Naming is a necessary contraction of a longer Functional Symbolic Trajectory. The contraction permits a community to carry extensive operations through a small number of marks. Someone who knows the word *logit* can traverse a large region of statistical practice without restating its history. Someone who does not know the word encounters a closed surface. The academic community can therefore be understood partly as a community possessing a shared compression dictionary. Its members have learned how to unfold specialised nouns into accepted operations, while an entering reader sees only the nouns.

The problem begins when the compressed name is treated as the origin rather than the terminus. The logistic function can then appear to be a mathematical thing waiting to be discovered, while the material and historical interactions that selected it recede from view. A problem was posed, values were registered, expressions were altered, computations were attempted, tables were used, rival procedures were compared, and professional communities accepted or resisted the result. The noun does not contain this history in an immediately visible form. It carries the authority produced by the history after the history itself has been removed from ordinary use.

There is therefore a difference between naming as compression and reification through forgotten compression. The first is unavoidable and productive. The second occurs when the symbolic remainder is granted a self-standing existence that obscures the operations necessary for its production. FSM does not seek to eliminate the noun. It seeks to restore the noun's provenance and to prevent a terminal symbolic state from being mistaken for a beginning.

The Historical FST of a Mathematical Object

The history of a mathematical operation can be represented as an ordered symbolic trajectory. Let $\mathcal{T}_H$ denote a finite historical FST assembled from surviving registrations such as texts, tables, diagrams, instruments, correspondence, software, and later accounts. Its schematic form may be written as

$$\mathcal{T}_H: h_1 \rightarrow h_2 \rightarrow h_3 \rightarrow \cdots \rightarrow h_K,$$

$$\text{problem} \rightarrow \text{proposed operation} \rightarrow \text{alternatives} \rightarrow \text{selection} \rightarrow \text{name} \rightarrow \text{canon.} \quad (3)$$

Here h_k is not an entire historical moment. It is a finite surviving registration admitted to the reconstruction, and K is the number of such registrations presently retained. The arrows declare order and provenance. They do not claim that history proceeded through a single unbranched line. On the contrary, the appearance of a line is itself a compression of a branching field of attempts, disappearances, rediscoveries, and retrospective classifications. The reconstruction remains conditional upon the surviving corpus.

An old printed volume is valuable within this inquiry because it arrests the symbolic trajectory at an earlier state. Later terminology has not yet been projected completely upon its pages. Operations may still be visible before they hardened into the names by which they are now taught. The page preserves notation, sentence structure, diagrams, ordering, and the assumed capacities of its intended reader. It is not an unmediated window upon the author's mind, but it is a surviving physical registration within the symbolic fossil record.

This is why Laplace, Recorde, Euclid, Newton, Greene, and other historical authors matter to FSM beyond the usual question of priority. The question is not merely who first discovered the object subsequently named. It is what operations were available, what distinctions were admitted, how words and marks were arranged, and what later descriptions did not yet exist. A modern textbook often projects its present nouns backwards, making the earlier writer look as though they were working towards the modern endpoint. The historical FST instead allows the earlier symbolic condition to retain its difference.

The word *fossil* must not imply immobility. Every surviving page enters new trajectories as it is read, translated, photographed, digitised, cited, and placed inside later arguments. The historical FST is therefore not a completed path recovered once and for all. It is a finite reconstruction formed from presently admitted traces. Its value is that it restores enough movement to challenge the apparent self-existence of the terminal noun.

The Reasonable Effectiveness of Surviving Mathematics

The celebrated question concerning the unreasonable effectiveness of mathematics is normally presented across a division. Mathematics is placed on one side as an abstract symbolic domain, while the physical world is placed on the other as the domain that mathematics mysteriously describes. The mystery depends upon the separation. If mathematical symbols developed independently of the interactions they later fit, their repeated success can appear miraculous.

The historical trajectory weakens that division. Mathematical operations are generated within physical and symbolic practices. Measurements are made; instruments are constructed; inscriptions are manipulated; results are compared with later registrations; inconvenient operations are altered; successful compressions are taught and repeated. The mathematics does not stand outside this loop. It is one of the finite physical-symbolic activities through which the loop proceeds.

The loop can be shown schematically as

$$\begin{array}{l} \text{interaction} \longrightarrow \text{registration} \longrightarrow \text{symbolic operation} \\ \longrightarrow \text{expectation} \longrightarrow \text{new interaction.} \end{array} \quad (4)$$

Each successful circulation alters the corpus from which a later operation can be selected. Procedures that repeatedly fail against new registrations may be abandoned, restricted, or revised. Procedures that permit stable manipulation and useful expectation remain available. Computational ease, institutional adoption, instrumentation, pedagogy, and software also participate in the selection. The surviving expression is therefore not a visitor from a separate mathematical realm. It is the terminal symbolic residue of many passages around the loop.

The apparent unreasonableness is intensified because the cemetery is missing. Modern presentation rarely displays every failed curve, discarded notation, inaccessible phenomenon, inconvenient calculation, or abandoned interpretation. It displays the survivors after selection has occurred. The reader sees a compact expression and the registered circumstances under which it works, but not the extensive population of symbolic alternatives through which the expression acquired its position.

We inherit the survivors and then ask why they have survived.

This does not make mathematical effectiveness trivial. The capacity to carry stable relations through very different registrations remains remarkable. The argument is narrower: effectiveness need not be unreasonable once the mathematics is returned to the historical and material trajectories through which it was produced, selected, and repeatedly adjusted. The mystery is partly manufactured by removing the provenance of the successful form.

II Before a Measurement Becomes a Statistic

The Generonic Boundary and Finite Registration

Statistics begins after a decisive physical event. Something has interacted with an instrument, a person, or another registration system strongly enough to produce a finite distinction. FSM calls the operational region through which this distinction is generated the generonic boundary. The boundary is not an extentless Euclidean line dividing a continuous world from a symbolic one. A detector integrates for a finite interval; a threshold has width, noise, and hysteresis; an observer requires time to discriminate one mark from another. Registration is an event with physical extent.

Let r_i denote the i th retained registration. A bare numerical record might carry only a value x_i , but a fuller registration may be represented by the tuple

$$r_i := (x_i, t_i, \Delta t_i, u_i, I_i, C_i, Q_i). \quad (5)$$

In this declaration, x_i is the registered value, t_i is its recorded time label, Δt_i is the relevant integration or sampling interval, u_i is the stated registration uncertainty, I_i records the instrument state, C_i carries admitted experimental conditions, and Q_i records relevant quality or processing information. No claim is made that every experiment must use precisely this tuple. Its purpose is to expose how much can disappear when the registration is reduced to a single magnitude.

The Alphonic Limit enters here as the commitment to a minimum presently distinguishable finite extent within a stated system of registration. It is not a timeless indivisible atom and not a thing-in-itself beyond measurement. It is a functional limit generated by an instrument, a handler, a corpus, and a consensus at a particular historical state. Two physical variations that do not produce distinguishable registrations at that limit enter the same symbolic state, even if a later or differently configured instrument separates them.

Every subsequent statistical operation acts downstream of this boundary. The statistician does not receive the unregistered interaction. The statistician receives symbols produced by a sequence of finite decisions, many of which may already have removed temporal, instrumental, and contextual relations. The data are consequently never raw. They are the output of earlier handlers, beginning with the physical-symbolic act of registration.

The Measured Functional Symbolic Trajectory

An ordered family of registrations forms the measured Functional Symbolic Trajectory. Let

$$\mathcal{T}_M := (r_1 \rightarrow r_2 \rightarrow \cdots \rightarrow r_N), \quad (6)$$

where N is the finite number of retained registrations. The arrow carries more than the fact that the values have been written from left to right. It declares that r_{i+1} follows r_i within the recorded history. If the intervals vary, that variation belongs to the trajectory. If an instrument changes state between two registrations, that change belongs to the trajectory. If a value was rejected and the rejection survives only as a gap, the gap belongs to the trajectory.

The distinction between a set of values and an FST is elementary but foundational. Consider the two sequences

$$\mathcal{T}_A := (1, 2, 3, 4), \quad \mathcal{T}_B := (4, 3, 2, 1). \quad (7)$$

They contain the same four magnitudes. A handler that retains only the arithmetic mean, the variance, or the unlabelled frequency of each value returns the same output for both. Yet $\mathcal{T}_A$ rises and $\mathcal{T}_B$ falls. Their successive differences have opposite signs. Their relation to any later state will differ if direction matters. They are not the same trajectory merely because a chosen operation does not distinguish them.

Long measured trajectories are especially vulnerable to contraction. The practical burden of retaining, presenting, and interpreting every registration encourages replacement by a distribution, a fitted curve, a coefficient, or a small collection of summary values. The long path then becomes a statistical object. The object may represent some relations extremely well, but it no longer contains the whole ordering from which it was made.

This loss is sometimes treated as though it arose naturally from the data. FSM instead places the handler inside the account. The trajectory did not transform itself into a mean or a histogram. A finite operation was performed, under a particular licence, for a stated or unstated purpose. To understand the result is therefore to understand both the measured FST and the operation applied to it.

Two Trajectories Beneath One Result

The historical FST and the measured FST approach the statistical result from different directions. The historical path generates the operation that can be selected. The measured path generates the registrations upon which the selected operation acts. Their meeting can be written as

$$\mathcal{T}_H \longrightarrow \boxed{\text{statistical method}} \longleftarrow \mathcal{T}_M. \quad (8)$$

The left-hand arrow carries the history of the method. It includes the problems for which the operation was proposed, the alternatives against which it was selected, the language in which it was described, and the computational and institutional circumstances of its survival. The right-hand arrow carries the history of the measured registrations. It includes order, intervals, uncertainty, instrument state, and the transformations performed before the operation was applied.

A published sentence such as ‘a logistic regression was performed’ compresses both sides. The word *logistic* names the historical survivor. The reported coefficients name the surviving output of the measured trajectory after the model has acted upon it. The sentence may be entirely adequate for a local scientific purpose, but its adequacy should not be confused with completeness. It presents the point at which two long trajectories have been contracted into a portable result.

The two contractions are structurally related because both map many earlier distinctions into fewer surviving symbols. They are not, however, immediately identical. Historical nounification is a corpus-level contraction whose admissible alternatives are often only partly recoverable. Statistical compression can be a formally declared operation upon a known finite sequence. Entropy will therefore be developed most directly on the measured side, where the handler and its preimage can be specified, while the historical side supplies the broader mechanics of forgotten provenance.

III The Statistical Handler

From Trajectory to Equivalence Class

Let $\mathcal{C}$ denote a finite symbolic handler acting upon an admitted family of measured trajectories. For a trajectory $\mathcal{T}$, the surviving statistical registration is declared by

$$s := \mathcal{C}(\mathcal{T}), \quad (9)$$

where s may be a number, a vector, a fitted curve, a table, a class label, or another finite symbolic result. The form of s matters, but the fundamental point is that s is an output of an operation. It is not the trajectory itself.

Suppose that two distinguishable trajectories $\mathcal{T}_A$ and $\mathcal{T}_B$ produce the same registered output under $\mathcal{C}$. We write

$$\mathcal{T}_A \sim_{\mathcal{C}} \mathcal{T}_B \iff \mathcal{C}(\mathcal{T}_A) = \mathcal{C}(\mathcal{T}_B). \quad (10)$$

The symbol $\sim_{\mathcal{C}}$ declares equivalence only under the named handler. It does not erase every other possible difference. A mean handler can make a rising and a falling trajectory equivalent while a delay handler separates them. A histogram handler can make a periodic sequence and its random permutation equivalent while a return map does not. Equivalence is therefore produced relative to an operation and a purpose.

For an output s , the associated trajectory class is

$$[s]_{\mathcal{C}} := \{\mathcal{T} \in \mathcal{A} : \mathcal{C}(\mathcal{T}) = s\}, \quad (11)$$

where $\mathcal{A}$ is the finite or finitely specified family of trajectories admitted to the investigation. The dependence upon $\mathcal{A}$ is essential. Without a declared basin of alternatives, the statement that many trajectories are compatible with s remains indefinite. A change in admissibility changes the class, even when the printed result s remains unchanged.

This construction reveals a statistic as a boundary-making operation. The handler retains distinctions that alter s and collapses distinctions that do not. The result is therefore not simply a smaller copy of the measured process. It is a new symbolic organisation in which a particular equivalence has been imposed.

Statistical equivalence is produced by an admissibility rule; it is not an identity already present in the measured trajectories.

Exchangeability and the Licence to Remove Order

Exchangeability provides a clear formal example of trajectory contraction. Let π denote a permutation of the indices $1, 2, \dots, N$. Applying π to an ordered sequence $\mathcal{T} = (x_1, x_2, \dots, x_N)$ produces

$$\pi\mathcal{T} := (x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}). \quad (12)$$

If a handler is invariant under the admitted permutations, then

$$\mathcal{C}(\mathcal{T}) = \mathcal{C}(\pi\mathcal{T}). \quad (13)$$

Equation (13) does not say that the permuted sequence is the same historical path. It states that the handler has been licensed not to register the difference. The operation passes from the ordered trajectory to an orbit of trajectories under permutation. Within that operation, order no longer contributes to the surviving result.

For N pairwise distinguishable registrations, removal of order permits as many as $N!$ distinct arrangements to enter the same unordered collection. If the retained magnitudes fall into k repeated symbolic states with multiplicities $n_1, n_2, \dots, n_k$, where the declared count satisfies $n_1 + n_2 + \dots + n_k = N$, the number of distinct orderings is

$$\Omega_{\text{ord}} := \frac{N!}{n_1! n_2! \dots n_k!}. \quad (14)$$

The quantity Ω_{ord} is a finite count generated under the stated symbolic distinctions. It does not assert that every ordering was physically possible or equally warranted. A rising temperature sequence and an arbitrary permutation may not be equally admissible under the conditions of the experiment. The factorial count exposes the multiplicity created by removal of order; provenance is still required to determine which members of that multiplicity belong to the admissible physical basin.

Exchangeability can therefore be read within FSM as a licence. It says that, for the stated operation, the order of specified registrations will not affect the result. This licence can be entirely appropriate. What matters is that the licence remains visible and is not later mistaken for evidence that no trajectory ever existed.

Different Statistics Retain Different Histories

It would be inaccurate to describe statistics as a single operation that invariably destroys trajectory. Statistical practices preserve different relations at different depths. A mean retains an aggregate magnitude while discarding order. A histogram retains frequencies within declared bins while discarding the sequence of visits to those bins. An autocorrelation retains average relations at selected lags but not the complete local path. A Markov model retains a declared depth of transition dependence. A state-space model introduces an explicit transition and observation structure, but still privileges the relations admitted by that structure.

The distinction can be summarised without treating the entries as exhaustive.

Handler	Relations retained	Relations commonly contracted
Arithmetic mean	Aggregate magnitude under a declared scale	Order, local transition, recurrence, and timing
Histogram	Frequency within stated bins	Sequence of visits, direction, and intervals
Variance	Dispersion about a declared centre	Local geometry and temporal direction
Correlation	Selected joint linear relation	Particular path and many non-linear relations
Autocorrelation	Average relation at selected lags	Individual return geometry and local transitions
Markov model	Transition dependence of stated order	History beyond the admitted state depth
Autoregressive model	Declared lagged linear dependence	Relations outside the chosen lag and model form
Logistic regression	Covariate–outcome relation under the selected link	Usually row order, temporal provenance, and unmodelled dependence
State-space model	Proposed transition and observation relations	Structures not admitted by the proposed states and equations

This variation suggests the idea of *retention depth*: the kind and extent of historical relation that remains available after an operation. Retention depth is not yet offered as a universal scalar. Order, interval, recurrence, spatial neighbourhood, instrument state, and semantic provenance are different relations and cannot be placed casually upon one line. The term instead directs attention to a question that conventional presentation can hide: what portion of the trajectory does this particular symbolic handler still carry?

Statistics becomes especially necessary where the full path was never registered, has become impractical to preserve, or cannot be communicated without contraction. It is a

disciplined technology for reasoning from what remains. FSM does not remove that role. It adds a requirement that the retained relations and the removed distinctions be made explicit wherever the provenance permits.

IV Entropy as Unresolved Trajectory

From Compression to Multiplicity

The broad operation developed so far can be written as

$$\text{trajectory} \longrightarrow \text{compression} \longrightarrow \text{equivalence class} \longrightarrow \text{unresolved multiplicity}. \quad (15)$$

The phrase *trajectory collapse* will name the many-to-one symbolic transition through which formerly distinguishable FSTs become equivalent under a handler. Collapse here does not mean that the past physical events are changed. It means that the surviving record can no longer separate the histories. The multiplicity belongs to the relation between the contracted symbol and the admissible paths beneath it.

Historical nounification is also a trajectory collapse in this broad sense. Many arguments, failed proposals, material constraints, and alternative notations are carried forward under one canonical name. Yet an entropy cannot be assigned merely by announcing that history has been forgotten. A quantitative entropy requires a specified family of alternatives and, where the alternatives are not equally weighted, a warranted distribution over them. The concept of trajectory collapse is therefore broader than the quantitative use of entropy.

Let s be the output of handler $\mathcal{C}$ and let $[s]_{\mathcal{C}}$ be its admissible trajectory class. If that class contains a finite number of equally weighted members, an elementary unresolved-trajectory entropy can be declared as

$$H_{\mathcal{C}}(s) := \log_b \text{card}([s]_{\mathcal{C}}), \quad (16)$$

where $\text{card}([s]_{\mathcal{C}})$ is the finite number of admitted trajectories compatible with s , and b is the declared logarithmic base. The base sets the reporting unit; it does not alter the underlying class. If the admissible trajectories carry weights p_i , indexed by $i = 1, 2, \dots, M$, a weighted form is

$$H_{\mathcal{C}}(s) := - \sum_{i=1}^M p_i \log_b p_i, \quad \sum_{i=1}^M p_i = 1. \quad (17)$$

The final condition in Equation (17) is a normalisation declared within the probability construction. It is not a claim that the weights exist independently of the procedure by which they were assigned. Their provenance remains part of the entropy statement.

An FSM Reading of Entropy

Entropy is often introduced through disorder, randomness, missing information, or the number of microscopic configurations compatible with a macroscopic state. FSM draws these presentations towards a common operational centre. A macroscopic or statistical registration is a surviving symbol. Multiple finer trajectories or configurations become unresolved beneath it because the active registration system does not distinguish them. Entropy concerns the multiplicity that remains after this symbolic contraction.

Entropy measures unresolved multiplicity among the admissible trajectories compatible with a surviving finite symbolic registration.

This formulation makes entropy relational. It does not belong to the printed value s alone. It depends upon the handler $\mathcal{C}$, the admissible family $\mathcal{A}$, the resolution at which registrations are distinguished, and any weights placed upon the compatible trajectories. A change in instrument, bin width, temporal resolution, model, or provenance can change the unresolved class even when the broader phenomenon under investigation appears unchanged.

The Alphonic commitment supplies a finite floor to the active distinctions. At a stated registration scale, two variations that do not produce distinguishable symbols occupy the same symbolic state. A later system may separate them, but the earlier record cannot be retroactively treated as though it contained the later distinction. Entropy is therefore epochal and instrumental without becoming arbitrary. It is tied to what could be registered, preserved, and admitted within a particular finite practice.

This account also changes the direction in which randomness is usually introduced. Instead of beginning with a system said to possess randomness and then attaching entropy, FSM can begin with a finite trajectory, follow the contractions applied to it, identify the histories made indistinguishable, and only then quantify the unresolved multiplicity. The direction becomes

finite trajectory $\longrightarrow$ symbolic contraction $\longrightarrow$ indistinguishable histories $\longrightarrow$ entropy.
(18)

This does not compel every thermodynamic or information-theoretic entropy to be reduced immediately to the loss of order in a list. It provides an FSM question that must be asked of each use: which finite distinctions have been registered, which have been collapsed, and what admissible multiplicity remains under the surviving symbol?

Four Locations of Uncertainty

Uncertainty is commonly carried forward as though it were a single quantity attached to a result. The present construction separates at least four locations at which uncertainty can enter. They can interact, but they should not be merged merely because a final report expresses them with one interval or error term.

Registration uncertainty concerns the production of r_i at the generonic boundary. It includes detector resolution, noise, calibration, integration time, threshold behaviour, and the finite capacity of the registration system to produce one distinguishable state rather than another. It is carried here by the term u_i in Equation (5), though in an actual experiment it may require several components rather than one scalar.

Trajectory uncertainty concerns the path after ordering, timing, or relational provenance has been removed. A collection of individually precise measurements can acquire extensive trajectory uncertainty when their sequence is discarded. This uncertainty was not necessarily present in the magnitude of any one registration. It was produced by a later handler that placed multiple orderings beneath the same result.

Model uncertainty concerns the selection of a symbolic family through which the registrations will be interpreted. Choosing a Gaussian form, a logistic link, a Markov depth, or a particular state-space structure determines which relations can be represented and which will fall outside the model. The uncertainty does not disappear when one model returns a compact set of fitted parameters. The parameters are conditional upon the selected form.

Historical uncertainty concerns the incomplete provenance of the operation itself. Surviving texts may not preserve all alternatives, instruments, conversations, or reasons for adoption. The canonical history can be a retrospective reconstruction organised by later nouns. Historical uncertainty therefore accompanies $\mathcal{T}_H$ even when the modern operation is formally specified.

The distinction can be displayed as

$$u_{\text{reg}} \not\equiv u_{\text{traj}} \not\equiv u_{\text{model}} \not\equiv u_{\text{hist}}, \quad (19)$$

where $\not\equiv$ indicates that the terms are not to be treated as the same symbolic source. They may be coupled in a final analysis, but their provenance should remain recoverable. The distinction is especially important for experiments in which highly stable instrumentation produces a long ordered record that is later reduced to a few statistics. Instrumental precision does not compensate for a destroyed trajectory.

Randomness After the Loss of History

A sequence can appear statistically unremarkable when its elements are considered separately and yet exhibit structure when their delayed relations are examined. A frequency distribution asks how many registrations occupied each declared state. An FST asks what followed what. A return map asks how the reconstructed trajectory crossed a section and where it returned. These are not competing answers to one unchanged question. They are different questions created by different handlers.

The word *random* can therefore refer to several distinct failures of discrimination. It may refer to an interaction for which no stable predictive relation has been registered. It may refer to a sequence that passes selected statistical tests. It may refer to the absence of a visible structure under a chosen embedding. It may refer to missing provenance that prevents one of several candidate trajectories from being preferred. Without the handler and the retained relations, the word compresses these different situations into one noun.

FSM does not require the declaration that all apparent randomness conceals a recoverable deterministic path. Such a declaration would exceed the registrations. It requires only that the claimed randomness be indexed to the finite procedure that produced it. Failure to find structure in a histogram is not failure to find structure in delay space. Failure to find structure at one sampling interval is not failure at every interval. Failure to distinguish candidate histories in the surviving record is not proof that no historical distinction occurred.

The appropriate movement is consequently from an unqualified noun towards an operational statement. Instead of saying only that a sequence is random, the report should state which finite tests, order relations, delays, resolutions, and admissibility rules were applied. Randomness then becomes a result with provenance rather than a property asserted beyond the available registration.

V Reconstructing the Retained Trajectory

The Fractional Sequence of π

The fractional digits of π provide an unusually clean symbolic sequence for examining the difference between distribution and trajectory. FSM does not require the completed infinite expansion as a physically available object. An investigation begins with a finite prefix generated by a declared procedure and stored in a finite medium. Let

$$\mathcal{T}_{\pi,N} := (d_1, d_2, \dots, d_N), \quad (20)$$

where d_j is the j th registered fractional digit in the selected base and N is the finite prefix length. The base, algorithm, precision checks, machine state, and retained output together form the relevant provenance.

The sequence is favourable because its order is preserved, its alphabet is finite, and long prefixes can be produced under stable symbolic rules. A digit-frequency handler may show that the counts of the available digit symbols are close over a sufficiently long prefix. That closeness is itself a registered result. It does not exhaust the relations in the ordered prefix. The delayed succession of digits remains available for a different operation.

Measured experimental sequences are usually less accommodating. Sampling intervals may vary, detector states may drift, registrations may be missing, and preliminary processing may remove relations before the sequence is stored. The contrast with $\mathcal{T}_{\pi,N}$ is therefore not that π is truly dynamical while measurement is merely statistical. It is that the finite digit prefix often arrives with exceptionally well-preserved order and generation provenance, whereas measured results are frequently narrowed before an embedding is attempted.

The visibility of structure also depends upon density. A long symbolic prefix can revisit neighbouring delay states many times. A short or sparsely sampled experiment may not provide enough recurrent registrations to reveal a stable geometry at the active resolution. This is not an argument against the experimental trajectory. It is a requirement that sequence length, sampling, uncertainty, and recurrence density be treated as part of the reconstruction rather than as external technical details.

Delay Embedding as a Finite Reconstruction

Let $x_1, x_2, \dots, x_N$ be an ordered scalar channel extracted from a measured FST. Choose a positive integer delay τ and a positive integer embedding dimension m . For each index j large enough to supply the required delayed registrations, define the delay vector

$$\mathbf{X}_j^{(m,\tau)} := (x_j, x_{j-\tau}, x_{j-2\tau}, \dots, x_{j-(m-1)\tau}). \quad (21)$$

The vectors $\mathbf{X}_j^{(m,\tau)}$ form a finite reconstructed trajectory in an m -coordinate symbolic space. The coordinates do not introduce new measurements. They reorganise relations already retained in the sequence. Each reconstructed point carries a present registration together with selected delayed registrations from its own provenance.

This construction is central to FSM because the geometry is produced by finite operations upon finite symbols. It need not be treated as proof of an inaccessible higher-dimensional object existing behind the sequence. The reconstruction is a measurable organisation of delayed relations. Its admissibility depends upon the quality and ordering of the retained registrations, the selected delay, the selected dimension, and the stability of the process during the recorded interval.

The parameters m and τ are handlers, not neutral windows. A delay that is too short may place successive coordinates so close that the reconstruction collapses towards a diagonal. A delay that is too long may separate registrations whose relevant coupling has already decayed. A dimension that is too small may fold distinguishable trajectories together, while a larger dimension may expose the sparsity and uncertainty of the available sequence. Parameter choice must therefore remain part of the provenance of every reconstructed geometry.

Where several measured channels are available, the FST need not be forced into one scalar stream without examination. A compound delay vector may include intensity, spatial position, detector state, or another registered channel. The admissible construction will depend upon which channels share a meaningful order and how their registration intervals are related. FSM does not grant higher dimension an independent ontology; it treats the coordinate array as an explicitly constructed symbolic organisation.

The Poincaré Section Is Not a Metaphor

Once an FST has been embedded, a section can be constructed through the reconstructed trajectory. Let $\mathcal{R}^{(m,\tau)}$ denote the finite ordered sequence of delay vectors from Equation (21), and let Σ denote a declared codimension-one section through the coordinate region occupied by those vectors. A crossing rule determines which passages through Σ are registered and in which direction.

If the successive registered crossings are

$$p_1, p_2, \dots, p_L, \quad p_\ell \in \Sigma, \quad (22)$$

then the finite return relation is declared by

$$P_\Sigma(p_\ell) := p_{\ell+1}, \quad \ell = 1, 2, \dots, L-1. \quad (23)$$

Here P_Σ is the Poincaré return handler associated with the constructed section, p_ℓ is the ℓ th admitted crossing, and L is the finite number of crossings retained. The construction proceeds through the full route

$$\text{registered interaction} \longrightarrow \text{FST} \longrightarrow \text{delay embedding} \longrightarrow \mathcal{R}^{(m,\tau)} \longrightarrow \Sigma \longrightarrow P_\Sigma. \quad (24)$$

This is not a figurative comparison between statistics and a Poincaré section. The section is a constructed geometrical solution within the reconstructed finite symbolic dynamics. Its crossings can be plotted, counted, compared, and revisited. The return map retains a relation between successive visits that a distribution of coordinate values may remove.

A distribution and a return map therefore occupy different operational positions. The distribution $p(x)$, where that notation has been warranted, records a weighting over declared symbolic states. The return map P_Σ records where an admitted crossing is followed by the next admitted crossing. One asks how registration is distributed; the other asks how the reconstructed trajectory returns. Neither replaces the other, but they do not preserve the same history.

The conventional statistic asks what remains after contraction. The Poincaré construction asks how the retained trajectory returns before that contraction is allowed to become terminal.

The Generonic Boundary and the Dynamical Separatrix

Two boundaries must now be kept distinct. The generonic boundary belongs to the production of finite symbols. It is the physical-symbolic region in which an interaction becomes a distinguishable registration. The dynamical separatrix belongs to the geometry reconstructed after a sequence of such registrations has been embedded. It separates neighbouring reconstructed paths that proceed towards different symbolic basins or stable modes.

The architecture is

$$\text{measured interaction} \xrightarrow{\mathcal{G}} \text{FST} \xrightarrow{(m,\tau)} \mathcal{R}^{(m,\tau)} \xrightarrow{\Sigma} P_{\Sigma} \longrightarrow \text{basins and separatrices.} \quad (25)$$

The symbol $\mathcal{G}$ denotes the declared generonic registration process, while (m, τ) denotes the embedding parameters. The separatrix is not used as a poetic name for the passage from world to symbol. It is sought within the reconstructed symbolic dynamics after that passage has produced an ordered FST.

This distinction allows the title *The Separatrix of the Symbol* to retain a precise meaning. The symbols are first generated through finite registration. Their ordered relations are then reconstructed. Only within that reconstruction can neighbouring paths, stable modes, transitions, and their separating regions be investigated. The separatrix is therefore an outcome of the operational geometry, not an assumed metaphysical boundary.

What Reconstruction Cannot Recover

Delay embedding does not reverse every compression. If only a mean, a histogram, or a vector of fitted coefficients survives, the temporal ordering from which the result was produced is generally not recoverable. The equivalence class may contain many admissible FSTs, and the statistic alone supplies no operation for selecting the one that occurred.

This limit can be written as

$$s = \mathcal{C}(\mathcal{T}) \not\Rightarrow \text{a unique } \mathcal{T}. \quad (26)$$

Equation (26) is not a statement that no useful inference is possible. Additional provenance, physical constraints, or retained partial order can reduce the admissible class. The statement is that the compressed result by itself does not warrant the restoration of distinctions removed by the handler.

The reconstruction also remains finite and conditional. A visible basin in one embedding can change when the delay, dimension, resolution, crossing rule, or sequence length changes. Such dependence does not make the geometry illusory. It makes the handler visible. The appropriate report includes these choices rather than allowing the image to present itself as an unconditioned portrait of the process.

The methodological consequence is direct.

Preserve first. Embed second. Compress later.

This order does not prohibit an experiment from using conventional statistics. It ensures that the statistical branch does not become the only surviving record before the FST has been examined.

VI An Operational Programme

Preserving the Experimental FST

An experimental record intended for FSM analysis should preserve the order and provenance required for later reconstruction. The primary record should be distinguished from every downstream representation. A practical primary registration might carry the form already introduced in Equation (5), extended where necessary to include spatial position, channel identity, transformation history, and links to calibration records.

The primary FST can then branch without being replaced by any one branch:

$$\mathcal{T}_M \longrightarrow \left\{ \begin{array}{l} \text{delay reconstruction,} \\ \text{Poincaré section and return map,} \\ \text{frequency distribution,} \\ \text{regression or fitted model,} \\ \text{summary statistics.} \end{array} \right. \quad (27)$$

The branches may be compared because they retain a common provenance. A histogram can be interpreted alongside a return map rather than in place of it. A fitted coefficient can be traced back to the registrations from which it was produced. If preprocessing changes the sequence, the transformed sequence becomes a new FST linked to its parent by an explicit handler.

Missing and rejected registrations require particular care. Deleting a value and closing the gap silently creates a false adjacency between the registrations on either side. Retaining a missing-state symbol preserves the fact that the trajectory passed through an unresolved interval. Similarly, smoothing, averaging, thresholding, and resampling create new symbolic trajectories. They may be useful, but their relation to the primary record must remain available.

The purpose of this discipline is not to accumulate unlimited metadata. Every experiment remains finite. The purpose is to decide, before contraction, which provenance is necessary to distinguish the trajectories relevant to the question. FSM makes that decision part of the experimental design rather than an afterthought performed upon a completed statistical table.

A Comparative Demonstration

Consider three finite sequences constructed from the same multiset of registered states:

$$\mathcal{T}_A := (1, 2, 3, 4, 3, 2, 1, 2, 3, 4, 3, 2), \quad (28)$$

$$\mathcal{T}_B := (4, 3, 2, 1, 2, 3, 4, 3, 2, 1, 2, 3), \quad (29)$$

$$\mathcal{T}_C := (2, 4, 1, 3, 2, 3, 1, 4, 3, 2, 3, 2). \quad (30)$$

The entries have been chosen so that simple frequency handlers can be made identical or very close after an appropriate finite extension, while the succession rules remain different. $\mathcal{T}_A$ contains a repeated rise and fall. $\mathcal{T}_B$ begins in the opposite phase. $\mathcal{T}_C$ disrupts the neighbouring order while retaining the same small alphabet and broadly similar counts.

A histogram handler maps the sequences towards the same frequency description. A one-step transition handler distinguishes how often each symbol follows another. A delay embedding with $m = 2$ and $\tau = 1$ turns each adjacent pair into a coordinate point. The rising and falling sequences then occupy structured paths through the coordinate plane, while the disrupted sequence visits the same broad region in a different order. A section through that plane records different returns even where the marginal distributions remain similar.

The demonstration is deliberately elementary. It shows that the same finite registrations can support several incompatible presentations without requiring large data or sophisticated probability. The histogram is not incorrect. It answers a frequency question. The embedding and return map answer trajectory questions that the histogram was not constructed to preserve.

The full monograph should eventually be accompanied by a reproducible computational supplement in which the sequences, handlers, embeddings, sections, and return maps are generated from one retained primary file. That supplement would allow the reader to vary the delay, dimension, binning, and crossing rule and to observe how each handler changes the surviving structure.

Towards the Winter Light Experiments

The argument acquires its physical importance in the planned investigation of light registration through pinholes, diffraction rings, and CCD detection. The purpose of the present monograph is not to settle the behaviour of light in advance. It is to establish the symbolic architecture through which the experiment can retain what it registers before those registrations are reduced to conventional image statistics.

The experimental route can be written as

$$\begin{aligned} \text{light interaction} &\longrightarrow \text{detector response} \longrightarrow \text{ADC registration} \\ &\longrightarrow \mathcal{T}_M \longrightarrow \text{embedding} \longrightarrow \Sigma \longrightarrow P_\Sigma. \end{aligned} \tag{31}$$

The analogue-to-digital converter occupies a decisive generonic position. It does not receive an already symbolic intensity. It integrates and discriminates an electrical response under finite thresholds, timing, bit depth, calibration, and noise. The pixel value is therefore an endpoint of an instrumental trajectory. A camera frame contracts many such endpoints into a spatial array, while a sequence of frames adds another temporal order.

Several candidate FSTs can be constructed from the same experiment. One may follow the temporal intensity of a selected pixel or region. Another may traverse the spatial registrations within a frame. A compound trajectory may carry radius, intensity, frame time, and detector state. These are not interchangeable orderings. Each creates a different delayed geometry and must be warranted by the physical and instrumental pathway under investigation.

The experimental design should therefore ask which primary ordering is produced by the apparatus, which orderings are later imposed by file format or scanning convention, and which transformations occur before the values become available. Dark-frame subtraction, averaging, clipping, interpolation, and image compression all create derived trajectories. None need be prohibited, but each should remain connected to the primary registrations through an explicit handler.

The winter experiment can then compare the conventional representation of diffraction rings with the embedded FST of the registered signal. The central question will not be whether one representation defeats the other. It will be whether stable return structures, basin changes, or separatrices are carried by the ordered registrations and lost when the frame is treated only as a static distribution of intensities.

VII The Place of Statistics Within FSM

Decompressing the Statistical Noun

The inquiry can now return to its opening words. To decompress a term such as *logit* is not to replace it with a longer definition and then continue as before. It is to recover the different trajectories concealed by its use. The historical FST explains how the operation entered the corpus. The formal handler explains what transformation is performed. The measured FST explains what registrations are supplied to it. The resulting equivalence class explains which distinct trajectories can produce the same output.

The direction of analysis is therefore

$$\boxed{\text{statistical noun}} \longrightarrow \text{historical operation} \longrightarrow \text{admissibility rule} \quad (32) \\ \longrightarrow \text{retained relations} \longrightarrow \text{collapsed trajectories.}$$

This direction reverses the usual pedagogical compression. The student normally receives the noun and learns how to reproduce its accepted operation. FSM asks what had to be removed for the noun to become stable, and what further removal occurs when that operation is applied to measured registrations.

The method can be applied beyond logit and probit. A Gaussian distribution, an Abelian group, Newton's second law, a Hilbert space, a confidence interval, and an entropy each carry a historical symbolic trajectory as well as a present operational definition. The point is not that all such terms are illegitimate. The point is that their legitimacy must arise from finite operations, provenance, and stated admissibility rather than from the authority of the noun alone.

Statistics as the Mathematics of What Remains

Statistics occupies a necessary place within FSM. A finite investigator cannot replay every physical interaction or carry every registration into every later statement. Some contraction is unavoidable. The statistical method provides disciplined ways to act after a part of the trajectory has become unavailable, impractical, or deliberately excluded from the question.

The change proposed here is one of position. The statistic should not be placed at the origin of the account or allowed to become the whole measured object. It is a downstream symbolic construction. Its authority is local to its handler, its admissibility conditions, and the provenance of the registrations upon which it acted.

An FSM statistical practice would therefore retain the primary FST where possible, declare the handler, identify the equivalence relation it creates, state which trajectory distinctions are removed, quantify unresolved multiplicity only where the admissible alternatives and weights have been warranted, and compare the contracted result with the reconstructed geometry carried by the ordered sequence. These are not separate bureaucratic requirements. They are parts of one continuous provenance.

The resulting position is neither anti-statistical nor a return to an imagined unmediated observation. Every FST has already passed through a generonic boundary. Every embedding is a handler. Every section is constructed. The advantage lies in keeping these operations visible and ordered. The statistical result can then be understood as one finite projection among others rather than as the process itself.

The Reader at the Terminal Section

A modern reader encounters the endpoint of both trajectories. The method arrives under a stable noun; the measurement arrives under a compressed result. The page appears to hold a completed object. Yet reading is itself another Functional Symbolic Trajectory. The marks are registered in sequence, measured against the reader's existing corpus, and unfolded into an interpretation that did not exist on the page alone.

The reader therefore becomes an author of the next trajectory. This does not mean that every reading is equally admissible. The printed symbols, their ordering, the surrounding corpus, and the reader's capacity to reconstruct the declared operations constrain what can be made. Meaning arises through that constrained interaction rather than being transported as an intact thing from one mind to another.

The complete movement can be written as

$$\begin{aligned} &\text{interaction} \longrightarrow \text{registration} \longrightarrow \text{FST} \longrightarrow \text{compression} \\ &\longrightarrow \text{published symbol} \longrightarrow \text{reader reconstruction.} \end{aligned} \tag{33}$$

The sentence 'a logistic regression was performed' can now be read differently. Beneath the word *logistic* lies the historical FST of the operation. Beneath the coefficients lies the measured FST and the handler that contracted it. Beneath the reader's understanding lies a further trajectory through a learned compression dictionary. What appeared to be a small declarative sentence is the intersection of several finite histories.

The monograph began before the numbers became statistics. It ends after the statistic has become another reader's registration. Between these points, no symbol stands alone. Every symbol is produced, carried, contracted, interpreted, and returned to a changing corpus. The statistic is not the end of the trajectory. It is a section through it.

Appendix A: Working Definitions

Admissible family.

The finite or finitely specified set $\mathcal{A}$ of trajectories permitted to enter a declared comparison or entropy construction.

Alphonic Limit.

The minimum presently distinguishable finite extent produced within a stated system of registration, instrument, handler, epoch, and consensus.

Dynamical separatrix.

A boundary within reconstructed symbolic dynamics separating paths that proceed into distinguishable basins or stable modes.

Finite registration.

A physically produced distinguishable symbolic state carrying uncertainty and provenance.

Functional Symbolic Trajectory.

An ordered finite pathway through which registrations, representations, constraints, uncertainty, and provenance are carried.

Generonic boundary.

The finite physical-symbolic region through which an interaction produces a distinguishable registration.

Historical FST.

A finite reconstruction of the ordered surviving registrations through which a symbolic method, noun, or corpus position was produced.

Measured FST.

The ordered sequence of finite registrations generated by an experiment, observation, computation, or other registered process.

Poincaré return handler.

The declared operation P_Σ that carries one admitted crossing of a constructed section Σ to the next admitted crossing.

Retention depth.

A provisional description of the kinds and extent of historical relations preserved by a handler.

Statistical handler.

A finite symbolic operation $\mathcal{C}$ that transforms an admitted trajectory into a statistical registration s .

Trajectory collapse.

A many-to-one symbolic operation through which distinguishable FSTs become equivalent under a specified handler.

Trajectory uncertainty.

Uncertainty concerning the generating path after ordering, timing, or relational provenance has been removed or contracted.

Unresolved multiplicity.

The admitted family of trajectories that remain compatible with a surviving symbolic result and are not distinguished by that result.

Appendix B: Notation

Symbol	Declared role
r_i	The i th retained finite registration.
x_i	The registered magnitude or selected scalar channel within r_i .
N	The finite number of registrations in a measured sequence.
$\mathcal{T}_H$	A reconstructed historical Functional Symbolic Trajectory.
$\mathcal{T}_M$	A measured Functional Symbolic Trajectory.
$\mathcal{G}$	The declared generonic registration process.
$\mathcal{C}$	A declared statistical or symbolic handler.
s	The surviving finite output of $\mathcal{C}$.
$\mathcal{A}$	The admitted family of candidate trajectories.
$[s]_{\mathcal{C}}$	The trajectories in $\mathcal{A}$ that produce output s under $\mathcal{C}$.
$H_{\mathcal{C}}(s)$	Unresolved-trajectory entropy relative to s , $\mathcal{C}$, and the admitted weighting.
m	The declared embedding dimension.
τ	The declared integer delay measured in registration steps.
$\mathbf{X}_j^{(m,\tau)}$	The delay vector at sequence index j .
$\mathcal{R}^{(m,\tau)}$	The finite reconstructed trajectory formed by the delay vectors.
Σ	A constructed section through the reconstructed coordinate region.
p_{ℓ}	The ℓ th admitted crossing of Σ .
P_{Σ}	The return handler from one admitted crossing to the next.

Appendix C: A Minimal Order-Loss Calculation

Suppose a finite record contains twelve registrations distributed among four symbolic values with multiplicities $n_1 = 2$, $n_2 = 4$, $n_3 = 4$, and $n_4 = 2$. A handler that retains only these multiplicities admits

$$\Omega_{\text{ord}} := \frac{12!}{2! 4! 4! 2!} = 207,900 \quad (34)$$

distinct orderings under the stated symbolic distinctions. The equals sign in this calculation reports identity of the finite arithmetic outputs under the declared factorial operation. It does not declare that all 207,900 trajectories were physically attainable in the process that produced the record.

If, for an illustrative symbolic calculation, all orderings are granted equal weight and the logarithmic base is 2, the unresolved-order entropy is

$$H_{\text{ord}} := \log_2(207,900) \approx 17.67 \text{ bits.} \quad (35)$$

The numerical value is conditional upon the equal-weight licence. Experimental provenance may eliminate most of the orderings or assign highly unequal weights. The calculation therefore demonstrates the multiplicity produced by an order-removing handler; it does not infer a physical probability distribution without further registrations.

Appendix D: A Primary Experimental Record

For the proposed CCD work, a primary record should preserve enough information to reconstruct the order in which finite values were produced. A possible row structure is

$$r_i := (i, t_i, f_i, \xi_i, \eta_i, x_i, y_i, u_i, I_i, D_i, C_i, Q_i), \quad (36)$$

where i is the registration index, t_i the recorded time, f_i the frame identity, ξ_i and η_i the spatial pixel coordinates, x_i the registered intensity, y_i an optional secondary channel, u_i the stated registration uncertainty, I_i the instrument state, D_i the relevant detector or ADC configuration, C_i the experimental condition, and Q_i the quality and processing record. The letters can be changed once the apparatus is specified; consistency and provenance matter more than these provisional labels.

The primary file should not silently replace original integer ADC registrations with normalised floating-point values. Normalisation, dark-frame subtraction, interpolation, smoothing, clipping, and spatial averaging should each produce a derived FST with a stated parent and transformation. Missing registrations should retain an explicit missing-state marker or interval record rather than creating false adjacency.

The record must also distinguish physical time order from storage order. Pixel arrays are commonly serialised by a file convention that may not correspond to a physical propagation trajectory. A spatial traversal chosen for an embedding is therefore a handler imposed after registration unless the detector architecture warrants that order. Temporal frame order, pixel readout order, radial order, and angular order are separate candidate trajectories and should not be merged without a declared bridge rule.

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